INCH-POUND
MIL-DTL-1183/8B
w/AMENDMENT 1
20 August 2019
SUPERSEDING
MIL-DTL-1183/8B
20 December 2016

DETAIL SPECIFICATION SHEET

FITTINGS, PIPE, CAST BRONZE, SILVER-BRAZING; REDUCING COUPLING, BUSHING, AND ADAPTER, PIPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification sheet and MIL-DTL-1183.

REQUIREMENTS:

1. Reducing coupling. Reducing couplings (see figure 1), except the 6- by 5-inch size, shall be Type A (streamlined interior). The 6- by 5-inch reducing coupling is Type B (interior not streamlined).

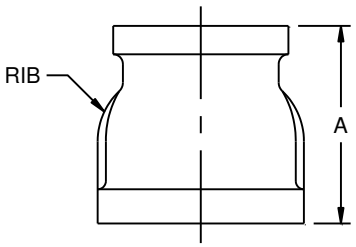


FIGURE 1. Reducing coupling.

2. Bushings. Bushings (male pipe-to-socket reducers) (see figure 2) shall be Type A (streamlined interior), which is achieved by tapering the interior to provide a gradual transition from the large diameter to the small diameter. Either method of transition specified on figure 2 is acceptable, provided the body wall thickness is not less than that specified in table V of MIL-DTL-1183 for the required end sizes.

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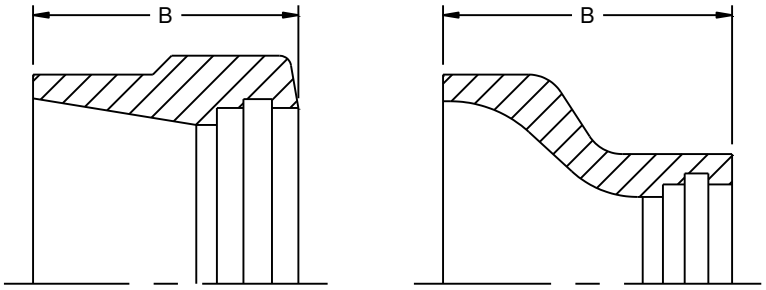


FIGURE 2. Bushing (male pipe to socket).

3. Adapters. Adapters (male pipe thread-to-socket reducers) (see figure 3) shall be Type B (interior not streamlined).

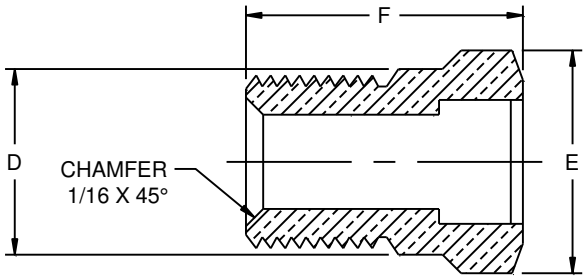


FIGURE 3. Adapter (male pipe thread to socket).

4. Dimensions. Dimensions for reducing couplings, bushings, and adapters (see figures 1, 2, and 3) shall be as specified in table I. The tolerance for the end-to-end (length) dimensions shall be twice the tolerance specified in table IV of MIL-DTL-1183 for the nominal pipe size of the reduced socket end.

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TABLE I. Dimensions, inches.

Nominal pipe size of larger socket (Figure 1)	Nominal pipe size of male pipe end (Figure 2)	Nominal pipe size male pipe thread end (Figure 3)	Nominal pipe size reduced socket end (Fig. 1, 2, or 3)	Reducing coupling length A	Bushing length B	Adapter		
						Min diameter D	Min band dia. E	Length F
⅜	⅜	⅜	¼	1.01	0.94	0.75	0.81	1.06
½	½		¼	1.13	1.06			
½	½	½	⅜	1.17	1.06	0.91	1.00	1.31
¾	¾		⅜	1.24	1.16			
¾	¾	¾	½	1.36	1.16	1.09	1.19	1.38
1	1		½	1.49	1.22			
1	1	1	¾	1.56	1.22	1.31	1.44	1.63
1¼	1¼		¾	1.65	1.44			
1¼	1¼	1¼	1	1.77	1.44	1.63	1.75	1.72
1½	1½		1	1.80	1.63			
1½	1½	1½	1¼	1.89	1.63	2.00	2.11	1.88
2	2		1¼	2.03	1.75			
2	2	2	1½	2.06	1.75	2.25	2.50	1.97
2½	2½		1½	3.25	1.94			
2½	2½	2½	2	3.25	1.94	2.81	3.03	2.63
3	3		2	3.69	2.13			
3	3		2½	3.69	2.13			
3½	3½		2½	4.00	2.19			
3½	3½		3	4.00	2.19			
4	4		3	4.38	2.25			
4	4		3½	4.38	2.25			
5	5		3½	5.00	2.75			
5	5		4	5.00	2.75			
6			4	5.75				
6			5	5.75				

5. Ribs. Reducing couplings, bushings, and adapters may be furnished with or without ribs.

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6. Specification sheet part number. The specification sheet part number is a definitive part number that is formed by combining the specification sheet code number (M1183/8 for MIL-DTL-1183/8) with a dash after it; the reducing couplings, bushings, adapters, and pipe size code number; the brazing ring groove and code letter; and an optional dash and lead-free fitting identifier. Reducers covered by this specification sheet shall be designated in the following form (see 6.2.b of MIL-DTL-1183):

M1183/8 – XX X - PW

Specification sheet code number

Reducing coupling/bushing/adapter style and pipe size code number (see requirement 7 herein)

Brazing ring groove and brazing ring code letter (see requirement 8 herein)

Lead-free fitting identifier (optional) (see 3.2 of MIL-DTL-1183)

7. Reducer style and pipe size. The reducing couplings, bushings, adapters, and nominal pipe sizes are identified by two digits (see table II).

TABLE II. Code number to style and size.

Reducer style			Nominal pipe size (inches)
1/ Reducing coupling (figure 1)	2/ Bushing (figure 2)	3/ Adapter (figure 3)	
01	41	81	⅜ x ¼
02	42		½ x ¼
03	43	82	½ x ⅜
04	44		¾ x ⅜
05	45	83	¾ x ½
06	46		1 x ½
07	47	84	1 x ¾
08	48		1¼ x ¾
09	49	85	1¼ x 1
10	50		1½ x 1
11	51	86	1½ x 1¼
12	52		2 x 1¼
13	53	87	2 x 1½
14	54		2½ x 1½
15	55	88	2½ x 2
16	56		3 x 2
17	57		3 x 2½
18	58		3½ x 2½
19	59		3½ x 3
20	60		4 x 3

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TABLE II. Code number to style and size – Continued.

Reducer style			Nominal pipe size (inches)
[1/] Reducing coupling (figure 1)	[2/] Bushing (figure 2)	[3/] Adapter (figure 3)	
21	61		4 x 3½
22	62		5 x 3½
23	63		5 x 4
24			6 x 4
25			6 x 5

NOTES:
[1/] Type A, except 6- by 5-inch size (see requirement 1).
[2/] Type A (see requirement 2).
[3/] Type B (see requirement 3).

8. Brazing ring groove and brazing ring. Face fed (no groove), grooved without brazing ring, grooved with brazing ring inserted in groove, grooved with brazing rings separately packaged and attached to the reducer, or grooved with brazing rings separately packaged and shipped with but not attached to the reducing couplings, bushings, and adapters socket ends are identified by a single letter (see table III).

TABLE III. Code letter to socket condition.

Code letter	Condition of socket
N	No groove (face fed)
G [1/]	Grooved (no brazing ring)
R [1/]	Grooved with brazing ring inserted
A [2/]	Grooved with brazing ring separately packaged and attached
S [1/ 2/]	Grooved with brazing ring separately packaged and shipped with

NOTES:
[1/] No National Stock Number (NSN) shall be established or exist in the Federal Supply System for elbow with the code letter of G, R, or S.
[2/] When the socket condition code letter is A or S, the brazing rings shall not be inserted in the grooves, but shall be packaged separately. For condition code letter A, the packaged brazing ring shall be either included within each individual package or physically attached to the reducing couplings, bushings, or adapters. For condition code letter S, the brazing ring shall be included within each individual package.

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CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH

Preparing activity:
Navy – SH
(Project 4730-2019-028)

Review activity:
DLA – CC

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